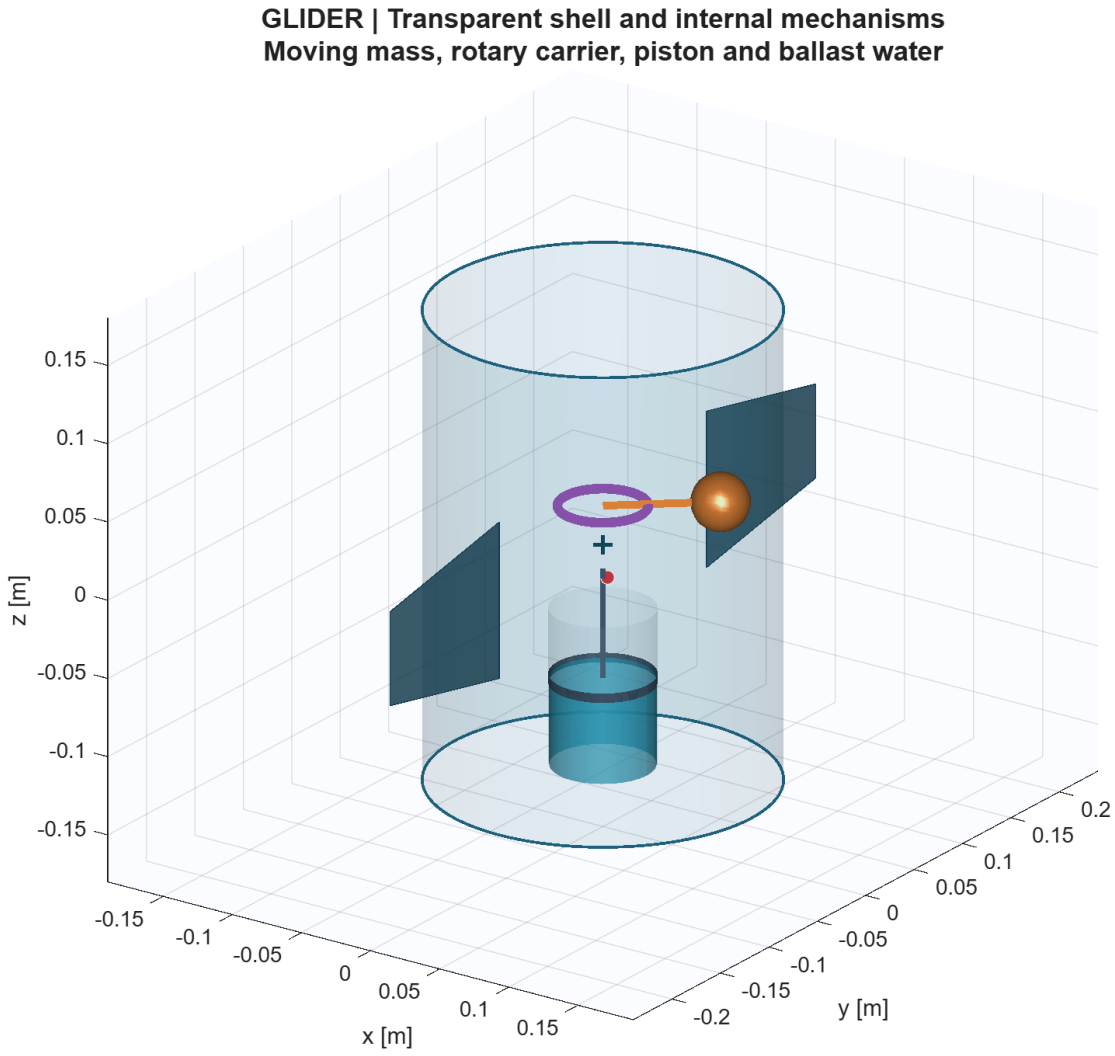
An underwater glider as a floating-base manipulator

Geometric modeling and simulation on SE(3) with variable ballast

Equations, MATLAB implementation and verification



Mechanism: a free base, a revolute–prismatic branch and a prismatic branch, with all joints enclosed by the hull.
Scope: an executable research model with explicitly identified demonstration parameters. Numerical verification does not establish experimental calibration.

Johnny Garcia and Julio October 26, 2026

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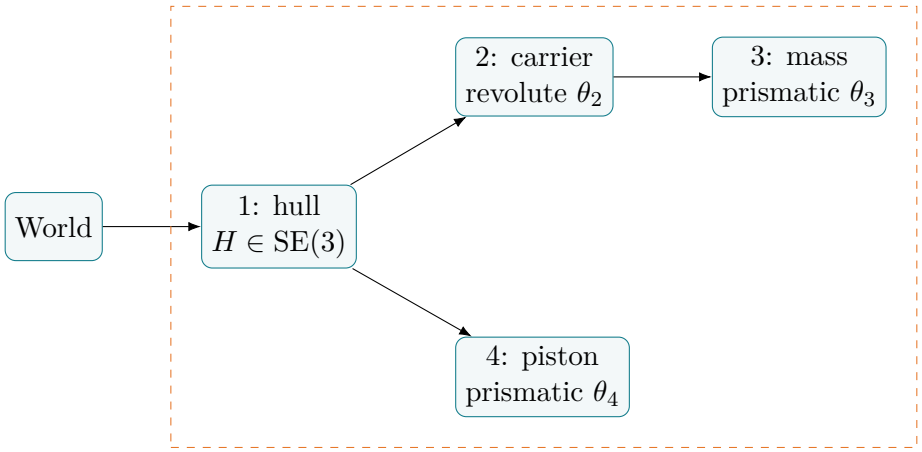
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1 Physical system and conventions

The hull is body 1 and encloses bodies 2, 3 and 4. One branch rotates a carrier and translates a mass; the other moves a syringe piston. Admitted water changes the total mass, its first moment and its inertia. The external displaced volume $\mathcal{V}$ remains constant. Hydrostatic buoyancy acts on the hull and is not assigned again to each internal body.



The model assumes still exterior water, uniform density, full submergence, rigid solids and no contact. Interior water is approximated by a distribution sharing the rigid velocity of the hull. Its relative circulation momentum is neglected, while momentum transport through the port is retained explicitly. The geometric center of the water is not treated as a material link.

The moving-mass and ballast models of Leonard and Graver motivate this physical separation [3, 4]. The spatial formulation of the solids adopts the angular-first convention of Modern Robotics [1]. Exterior-fluid added inertia is separated from dissipative forces, as in the structure of Fossen's marine craft model [2].

The world z-axis points upward, with $g_0 = (0, 0, -g)^\top$. The nominal longitudinal hull axis is z_B, retaining the supplied code's axis orientation. The NED convention is not used.

Symbol	Definition
$H = {}^0H_1 = [R, p; 0, 1]$	Pose mapping base coordinates to world coordinates.
$q = (\theta_2, \theta_3, \theta_4)^\top$	Internal coordinates, in radians and meters.
$V = (\omega^\top, v^\top)^\top \in \mathbb{R}^6$	Base twist expressed in the base frame.
$\nu = (V^\top, \dot{q}^\top)^\top \in \mathbb{R}^9$	Generalized velocity, not nine Cartesian coordinate derivatives.
$W = (n^\top, f^\top)^\top$	Wrench, moment first; power is $V^\top W$.
$\mathsf{E} = [I_6 \;\; 0]$, $\mathsf{B} = [0 \;\; I_3]$	Selectors: $V = \mathsf{E}\nu$, $\dot{q} = \mathsf{B}\nu$.

The configuration space is $\mathrm{SE}(3) \times S^1 \times \mathbb{R}^2$, restricted by physical travel limits. The angle is unwrapped within $[-\pi, \pi]$. Crossing a limit stops the simulation without representing an impact.

2 Geometric tree kinematics

For $x \in \mathbb{R}^3$, let $[x]_\times y = x \times y$. In angular-first order,

$$\widehat{V} = \begin{bmatrix} [\omega]_\times & v \\ 0 & 0 \end{bmatrix}, \quad \mathrm{Ad}_H = \begin{bmatrix} R & 0 \\ [p]_\times R & R \end{bmatrix}, \quad \mathrm{ad}_V = \begin{bmatrix} [\omega]_\times & 0 \\ [v]_\times & [\omega]_\times \end{bmatrix}. \tag{1}$$

The base satisfies $\dot{H} = H\widehat{V}$, allowing updates through the group exponential. The parents are $a(2) = 1$, $a(3) = 2$, $a(4) = 1$. The constant body screw axes are

$$S_2 = (0, 0, 1, 0, 0, 0)^\top, \quad S_3 = (0, 0, 0, 1, 0, 0)^\top, \quad S_4 = (0, 0, 0, 0, 0, -1)^\top. \tag{2}$$

Relative poses and velocity transformations are

$$H_i = {}^{a(i)}H_i(0)\exp(\widehat{S}_i\theta_i), \qquad X_i = \mathrm{Ad}_{H_i^{-1}}. \tag{3}$$

Let V_i be each body's twist relative to the world, expressed in its own frame. The recursions are

$$V_1 = V, \qquad V_i = X_i V_{a(i)} + S_i\dot{\theta}_i, \tag{4}$$

$$\dot{V}_1 = \dot{V}, \qquad \dot{V}_i = X_i\dot{V}_{a(i)} + S_i\ddot{\theta}_i - \mathrm{ad}_{S_i\dot{\theta}_i} X_i V_{a(i)}. \tag{5}$$

The Jacobians $J_i \in \mathbb{R}^{6\times 9}$ and bias accelerations a_i^0 are computed together:

$$V_i = J_i\nu, \qquad \dot{V}_i = J_i\dot{\nu} + a_i^0, \qquad J = \begin{bmatrix} I & 0 & 0 & 0 \\ X_2 & S_2 & 0 & 0 \\ X_3X_2 & X_3S_2 & S_3 & 0 \\ X_4 & 0 & 0 & S_4 \end{bmatrix}. \tag{6}$$

The revolute column for body 3 is X_3S_2, expressed in body 3.

2.1 Equivalent matrix form

Define

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ X_2 & 0 & 0 & 0 \\ 0 & X_3 & 0 & 0 \\ X_4 & 0 & 0 & 0 \end{bmatrix}, \quad L = (I - A)^{-1}, \quad \mathsf{S} = \mathrm{diag}(I_6, S_2, S_3, S_4). \tag{7}$$

The rectangular matrix S has size 24×9, and

$$J = L\mathsf{S}, \quad a^0 = -LDAJ\nu, \quad D = \mathrm{diag}(0_6, \mathrm{ad}_{S_2\dot{\theta}_2}, \mathrm{ad}_{S_3\dot{\theta}_3}, \mathrm{ad}_{S_4\dot{\theta}_4}). \tag{8}$$

The blocks of D lie on its diagonal. The recursive implementation avoids constructing or inverting the 24×24 stacked matrices.

3 Inertia, gravity and buoyancy

For mass $m_i > 0$, center of mass c_i expressed in the body frame, and central rotational inertia $J_{c,i}$, the spatial inertia is

$$\mathcal{I}_i = \begin{bmatrix} J_{c,i} - m_i[c_i]_\times^2 & m_i[c_i]_\times \\ -m_i[c_i]_\times & m_i I_3 \end{bmatrix}. \tag{9}$$

If $V_b = XV_a$, preservation of kinetic energy requires $\mathcal{I}_a = X^\top \mathcal{I}_b X$. A frame change transforms the complete inertia matrix through this congruence. The inertial wrench is

$$W_i^{\text{in}} = \mathcal{I}_i \dot{V}_i - \text{ad}_{V_i}^\top \mathcal{I}_i V_i. \tag{10}$$

With $R_i = {}^0R_i$, gravity and its generalized projection are

$$f_{g,i} = m_i R_i^\top g_0, \quad W_{g,i} = \begin{bmatrix} c_i \times f_{g,i} \\ f_{g,i} \end{bmatrix}, \qquad Q_{g,d} = \sum_{i=1}^{4} J_i^\top W_{g,i}. \tag{11}$$

All nine projected components are retained, including the joint components. Gravity and inertia use the same center-of-mass locations.

For center of buoyancy c_B expressed in the base,

$$f_B = -\rho \mathcal{V} R^\top g_0, \qquad W_B = \begin{bmatrix} c_B \times f_B \\ f_B \end{bmatrix}. \tag{12}$$

The minus sign produces an upward force using the same gravity vector throughout the derivation. Its projection is $\mathsf{E}^\top W_B$.

3.1 Center of mass and density

Let $r_i(q)$ denote each solid's center of mass in the base frame. For stored water mass m_w and center c_w,

$$m_{\text{tot}} = \sum_i m_i + m_w, \qquad c_G(q) = \frac{\sum_i m_i r_i(q) + m_w c_w}{m_{\text{tot}}}. \tag{13}$$

The mean density is $m_{\text{tot}}/\mathcal{V}$. Moving internal masses changes the center of mass and inertial couplings; pumping water also changes the total mass. Neutral buoyancy requires $m_{\text{tot}} = \rho \mathcal{V}$.

3.2 Physical admissibility

The model constructor checks symmetry, positive definiteness and the triangle inequalities for each central inertia:

$$\lambda_{\max}(J_c) \le \tfrac{1}{2} \operatorname{tr}(J_c). \tag{14}$$

A positive spatial matrix alone does not guarantee a physically realizable mass distribution. The supplied hull inertia violates this condition and is replaced by an explicitly provisional equivalent cylinder.

4 Stored water: geometry and variable inertia

The supplied code's convention is adopted: $0 \le \theta_4 \le \ell$, $\ell = 0.1\,\mathrm{m}$, with the piston moving along $-z_B$.

$$h = \ell - \theta_4, \quad A_s = \pi r_s^2, \quad m_w = \rho A_s h, \quad \dot{m}_w = -\rho A_s \dot{\theta}_4. \tag{15}$$

The chamber is full at $\theta_4 = 0$ and empty at $\theta_4 = \ell$. For a fixed lower end r_t and $e_3 = (0,0,1)^\top$,

$$r_p = r_t + h e_3, \qquad c_w = r_t + \tfrac{1}{2} h e_3. \tag{16}$$

A prime denotes differentiation with respect to θ_4: $r_p' = -e_3$ and $c_w' = -e_3/2$. Thus the geometric water center does not move with the piston. Frame 4 remains attached to the dry piston. For a uniformly filled cylinder,

$$J_{c,w} = \frac{m_w}{12}\,\mathrm{diag}(3r_s^2 + h^2, 3r_s^2 + h^2, 6r_s^2), \qquad \mathcal{I}_w = \mathcal{I}(m_w, c_w, J_{c,w}). \tag{17}$$

Let $k = \rho A_s$, $C = [c_w]_\times$, $C' = [-e_3/2]_\times$, $m_w' = -k$.

$$J_{c,w}' = -\frac{k}{12}\,\mathrm{diag}(3r_s^2 + h^2, 3r_s^2 + h^2, 6r_s^2) + \frac{m_w}{12}\,\mathrm{diag}(-2h, -2h, 0), \tag{18}$$

$$\mathcal{I}_w' = \begin{bmatrix} J_{c,w}' - m_w' C^2 - m_w(C'C + CC') & m_w' C + m_w C' \\ -m_w' C - m_w C' & m_w' I_3 \end{bmatrix}, \tag{19}$$

$$\dot{\mathcal{I}}_w = \mathcal{I}_w' \dot{\theta}_4. \tag{20}$$

At empty fill, $m_w = 0$ and $\mathcal{I}_w = 0$, without singularities. The water cylinder's rotational inertia is retained rather than replacing the stored volume by a point mass.

4.1 Interior-flow approximation

The occupied domain changes. The velocity of its geometric center does not determine particle velocities. In general,

$$p_w = \mathcal{I}_w V + p_{\mathrm{rel}}, \qquad p_{\mathrm{rel}} = \int_{\Omega_w} \begin{bmatrix} r \times u_{\mathrm{rel}}(r) \\ u_{\mathrm{rel}}(r) \end{bmatrix} dm. \tag{21}$$

The implemented closure takes $p_{\mathrm{rel}} \simeq 0$. This must be revisited when circulation, flow rate, sloshing or pipe flow produces appreciable relative momentum. A fill-dependent inertia alone then cannot replace an interior-flow model.

5 Open-system momentum balance

The momentum expressed in the base and referred to its origin is

$$P_B = \sum_i \mathrm{Ad}^{\top}_{1_{H_i^{-1}}} \mathcal{I}_i V_i + \mathcal{I}_w V + M_A V. \tag{22}$$

Here M_A is a constant, symmetric positive-semidefinite added hydrodynamic inertia of the exterior water; it is distinct from stored ballast water. For a control volume admitting or expelling water,

$$\dot{P}_B - \mathrm{ad}^{\top}_V P_B = W_{\mathrm{ext}} + \Phi, \qquad \Phi = \dot{m}_w \begin{bmatrix} r_o \times u_o \\ u_o \end{bmatrix}. \tag{23}$$

The effective port location is r_o, and u_o is the absolute transport velocity expressed in the base. The demonstration closure, without a resolved directional jet, is

$$u_o = \begin{cases} 0, & \dot{m}_w \geq 0 \quad \text{(inflow from still water)}, \\ v + \omega \times r_o, & \dot{m}_w < 0 \quad \text{(comoving outflow)}. \end{cases} \tag{24}$$

A real jet requires a consistent addition of relative velocity and pressure forces. Exterior current also requires changes to hydrodynamic forces and relative added-inertia dynamics. The option `water.flux='zero'` selects an alternative idealization of zero net momentum transport, requiring its own physical justification.

The stored-water inertial contribution is

$$\boxed{\mathcal{I}_w \dot{V} + \dot{\mathcal{I}}_w V - \mathrm{ad}^{\top}_V \mathcal{I}_w V.} \tag{25}$$

The inertia derivative has a positive sign on the inertial side; Φ is on the force side. Subtracting an inertia derivative from ad_V would combine objects of different physical dimensions and does not represent this balance.

5.1 Translational sign check

For a point mass without rotation,

$$m\dot{v} + \dot{m}\, v = F + \dot{m}\, u_o. \tag{26}$$

During admission of still water, $u_o = 0$: the vehicle accelerates the mass it acquires. During comoving expulsion, $u_o = v$, and the two $\dot{m}\, v$ terms cancel. Omitting flux cannot represent both processes indiscriminately.

The water contributes the gravity wrench

$$W_{g,w} = \begin{bmatrix} c_w \times (m_w R^{\top} g_0) \\ m_w R^{\top} g_0 \end{bmatrix}, \tag{27}$$

which enters the base rows under this closure. The hydraulic piston load is specified separately.

6 Hull, wings and syringe load

At a load center r_H, let $v_H = v + \omega \times r_H$, $s = \|v_H\|$, $e = v_H/s$, with unit wing normal n and unit lateral direction b_s. For $s > 0$, the demonstration force model is

$$f_L = -k_L s (n^\top v_H)[n - (n^\top e)e], \tag{28}$$

$$f_D = -[k_D + k_{D\alpha}(n^\top e)^2] s\, v_H - k_s |b_s^\top v_H| (b_s^\top v_H) b_s, \tag{29}$$

$$n_D = -d_\omega \odot \omega - k_\omega \odot |\omega| \odot \omega, \tag{30}$$

$$W_H = \begin{bmatrix} r_H \times (f_L + f_D) + n_D \\ f_L + f_D \end{bmatrix}. \tag{31}$$

At zero speed, $f_L = f_D = 0$. Dissipation coefficients are nonnegative. The coefficients $k_L, k_D, k_{D\alpha}, k_s$ have units kg/m; rotational coefficients produce n_D in N m. The power identities are

$$v_H^\top f_L = 0, \qquad V^\top W_H = v_H^\top f_D + \omega^\top n_D \le 0. \tag{32}$$

They hold for reversed flow as well. Lift contributes no instantaneous power. This force surrogate has neither experimental identification nor a stall model. Added inertia contributes $M_A \dot{V} - \mathrm{ad}_V^\top M_A V$. Fossen's linear-first matrices must be permuted before use with this project's angular-first convention.

6.1 Quasistatic hydraulic load

Pressure produces an equivalent internal piston effort:

$$x_p = p + R r_p, \quad \Delta p = p_{\text{surface}} + \rho g_0^\top x_p - p_{\text{back}}, \tag{33}$$

$$f_p = -A_s \Delta p - d_p \dot{\theta}_4 - f_c \tanh(\dot{\theta}_4 / v_s), \tag{34}$$

$$Q_p = (0, \ldots, 0, f_p)^\top \in \mathbb{R}^9. \tag{35}$$

Exterior pressure increases with depth. For $\Delta p > 0$, the negative load f_p opposes water expulsion, $\dot{\theta}_4 > 0$. The internal force and reaction cancel under a rigid variation of the complete assembly, so Q_p has no base component.

The example uses constant $p_{\text{back}} = p_{\text{surface}} = 101325\,\text{Pa}$, with $d_p = f_c = 0$. This is a constant-back-pressure assumption. A sealed gas chamber requires a pressure–volume law; valves, seals, pipe losses and efficiencies require further data. Setting `pump.enabled=false` disables this equivalent load.

7 Complete equations and adapted RNEA

The dry-body contribution is

$$M_d = \sum_i J_i^\top \mathcal{I}_i J_i, \tag{36}$$

$$b_d = \sum_i J_i^\top (\mathcal{I}_i a_i^0 - \mathrm{ad}_{V_i}^\top \mathcal{I}_i V_i). \tag{37}$$

The implemented system is

$$\boxed{M(q)\dot{\nu} + b(q,\nu) = \mathsf{B}^\top \tau + Q(H,q,\nu),} \tag{38}$$

where

$$M = M_d + \mathsf{E}^\top (\mathcal{I}_w + M_A)\mathsf{E}, \tag{39}$$

$$b = b_d + \mathsf{E}^\top [\dot{\mathcal{I}}_w V - \mathrm{ad}_V^\top (\mathcal{I}_w + M_A)V], \tag{40}$$

$$Q = Q_{g,d} + \mathsf{E}^\top (W_{g,w} + W_B + W_H + \Phi) + Q_p. \tag{41}$$

Actuation $\tau \in \mathbb{R}^3$ contains one torque and two forces. The base has no direct actuation, while external loads act on it. For constant solid-body inertias, an equivalent matrix expression is

$$C_d \nu = J^\top [-\mathcal{I} L D A - \mathrm{ad}_{\mathcal{V}}^\top \mathcal{I}] J\nu, \tag{42}$$

with $\mathcal{V} = J\nu$, $\mathcal{I} = \mathrm{diag}(\mathcal{I}_1, \ldots, \mathcal{I}_4)$ and $\mathrm{ad}_{\mathcal{V}} = \mathrm{diag}(\mathrm{ad}_{V_1}, \ldots, \mathrm{ad}_{V_4})$. Simulation needs the bias vector b; a unique Coriolis matrix need not be selected.

7.1 Recursive Newton–Euler passes

The forward pass obtains $V_i, \dot{V}_i$ using (5) and initializes

$$F_i = \mathcal{I}_i \dot{V}_i - \mathrm{ad}_{V_i}^\top \mathcal{I}_i V_i - W_{g,i}. \tag{43}$$

At the root, add

$$(\mathcal{I}_w + M_A)\dot{V} + \dot{\mathcal{I}}_w V - \mathrm{ad}_V^\top (\mathcal{I}_w + M_A)V - W_{g,w} - W_B - W_H - \Phi. \tag{44}$$

The backward pass processes bodies $4, 3, 2$:

$$\tau_i^* = S_i^\top F_i, \qquad F_{a(i)} \leftarrow F_{a(i)} + X_i^\top F_i. \tag{45}$$

Subtract Q_p from $(F_1, \tau_2^*, \tau_3^*, \tau_4^*)$. The resulting vector equals $M\dot{\nu} + b - Q$. Its base block is the wrench required by the requested acceleration and must vanish for a realizable unactuated-base acceleration.

The function `inverse_dynamics.m` executes this backward pass without calling the matrix assembly. The equivalence test checks both routes independently while sharing the kinematic formulation.

8 Simulation and integration on the group

8.1 Prescribed-joint mode

Given $q, \dot{q}, \ddot{q}$, partition the equations into base and joint blocks:

$$\dot{V} = M_{bb}^{-1}(Q_b - b_b - M_{bq}\ddot{q}), \tag{46}$$

$$\tau = M_{qb}\dot{V} + M_{qq}\ddot{q} + b_q - Q_q. \tag{47}$$

The implementation solves a linear system with backslash rather than forming an inverse. RNEA recovers the reactions. Sufficient actuation is assumed; a prescribed reference does not establish that a particular motor can realize it.

8.2 Effort-driven mode

For $\tau = \tau(t, H, q, \nu)$, solve

$$\dot{\nu} = M^{-1}(\mathsf{B}^\top \tau + Q - b), \qquad \dot{q} = \mathsf{B}\nu, \qquad \dot{H} = H\widehat{\mathsf{E}\nu}. \tag{48}$$

A callback can supply a controller. Travel limits stop integration; hard stops and impacts are not modeled.

8.3 Geometric midpoint method

Let $f(t, H, q, \nu)$ be the acceleration of the selected mode. For step Δt, with $a_n = \dot{\nu}_n$,

$$H_{1/2} = H_n \exp(\tfrac{\Delta t}{2}\widehat{V}_n), \qquad \nu_{1/2} = \nu_n + \tfrac{\Delta t}{2} a_n, \tag{49}$$

$$q_{1/2} = q_n + \tfrac{\Delta t}{2}\dot{q}_n, \qquad a_{1/2} = f(t_n + \tfrac{\Delta t}{2}, H_{1/2}, q_{1/2}, \nu_{1/2}), \tag{50}$$

$$H_{n+1} = H_n \exp(\Delta t \widehat{V}_{1/2}), \qquad \nu_{n+1} = \nu_n + \Delta t\, a_{1/2}, \tag{51}$$

$$q_{n+1} = q_n + \Delta t\, \dot{q}_{1/2}. \tag{52}$$

Prescribed joint references are evaluated directly at every stage. The method is second order for sufficiently regular dynamics and retains $H \in \mathrm{SE}(3)$ up to roundoff. It is neither exactly energy preserving nor a variational integrator.

The exponential uses Rodrigues coefficients and the translation Jacobian, with series near zero angle. It is checked against `expm`. The simulated pose is not integrated through Euler angles or projected afterward to repair orthogonality. The graphics transform alone is reconstructed to satisfy the renderer's numerical tolerances.

8.4 Example maneuver

Each transition uses $s(u) = 10u^3 - 15u^4 + 6u^5$, $0 \le u \le 1$, with analytic derivatives. The example alternates $\theta_4 = 0.015$ and $0.095\,\mathrm{m}$, holds $\theta_3 = 0.030\,\mathrm{m}$, and rotates the mass through $0, \pi, 0, -\pi, 0$. Transitions last four seconds, with sixteen-second constant segments.

9 Open-system energy balance

The mechanical energy stored by the reduced model is

$$K = \tfrac{1}{2}\nu^\top M\nu, \tag{53}$$

$$U = -\sum_i m_i g_0^\top (p + Rr_i) - m_w g_0^\top (p + Rc_w) + \rho\mathcal{V} g_0^\top (p + Rc_B), \tag{54}$$

$$E = K + U. \tag{55}$$

Let $s_w = m_w c_w$ be the water's first moment in the base. The balance is

$$\boxed{\dot{E} = \tau^\top \dot{q} + V^\top W_H + \nu^\top Q_p + P_{\text{open}},} \tag{56}$$

where

$$P_{\text{open}} = V^\top \Phi - \tfrac{1}{2} V^\top \dot{\mathcal{I}}_w V - g_0^\top (\dot{m}_w p + R\dot{s}_w). \tag{57}$$

Multiplying (38) by $\nu^\top$, the dry contribution satisfies $\nu^\top b_d = \frac{1}{2}\nu^\top \dot{M}_d \nu$. Coadjoint terms contribute no power. The derivative of K contains $\frac{1}{2} V^\top \dot{\mathcal{I}}_w V$, whereas the bias contains $V^\top \dot{\mathcal{I}}_w V$. Their difference gives the sign in (57). Conservative forces cancel the rigid-motion part of $\dot{U}$; changes in mass and first moment leave the final term.

At fixed fill, with no hydraulic load or dissipation, $\dot{E} = \tau^\top \dot{q}$. Pumping makes this an open system, so constant E is not expected. Four power channels are logged: actuation, hydrodynamics, transport and storage change, and hydraulic loading.

9.1 Physical interpretation and scope

The term $\nu^\top Q_p$ represents work exchanged with the fluid and rear chamber under the pressure closure. Gas internal energy and electrical losses are not included in E. Joint efforts alone do not determine electrical consumption. Transported potential power depends on the potential datum, as expected when mass crosses the control boundary.

Verification compares Jacobians with pose differences, inertia and bias with Euler–Poincaré/Lagrange calculations, gravity with potential derivatives, and RNEA with the matrix equations. It also checks mass flux, pressure signs and the instantaneous energy balance. Without external forces and with fixed ballast, the spatial momentum $\mathrm{Ad}_{H^{-1}}^\top P_B$ converges to a constant as the step is refined. A norm combining angular and linear momentum is a mixed-unit numerical indicator, not a universal physical error bound.

10 Parameters and provenance

Quantity	Value	Treatment
ρ, g	1000; 9.8	Supplied code, SI units.
Diameter and length	0.2142; 0.300 m	Nominal code dimensions.
Dry masses	10.780; .50003; .50003; .150 kg	Retained.
Displaced mass	12.096 kg	Effective volume $0.012096\,\mathrm{m}^3$.
Syringe	$d_s = .065,\ \ell = .100$ m	Supplied-code convention.
Travel limits	$\pm\pi$; $[0, .04]$; $[0, .1]$	Retained, radians and meters.
Dry center of body 1	$(0, 0, -.0225)$ m	Supplied gravity offset.
Inertias of 2, 3, 4	See `parameters.m`	Retained diagonal tensors.
Central inertia of 1	Equivalent cylinder	Admissible provisional replacement.
R mount and P guide	$(0, 0, .025)$; .040 m	Proposed enclosed mounting.
Syringe end	$(0, 0, -.140)$ m	Proposed axial clearance.
M_A	$\mathrm{diag}(.02, .02, .01, 2, 6, 3)$	Demonstration values: $\mathrm{kg\,m}^2$, then kg.
Hydrodynamics	Equation (31)	New, unidentified force law.
Back pressure	101325 Pa	Constant example assumption.

The nominal cylinder displaces approximately 10.81 liters, unlike the 12.096-liter effective volume used for buoyancy. Visual geometry does not determine that volume; a measurement of the complete assembly and appendages is still needed.

The original base inertia has $J_z > J_x + J_y$, even before resolving its center offset. The original moving mass can reach radius 0.13 m, exceeding 0.1071 m. The proposed layout leaves 9.1 mm radial clearance for the visual mass sphere over its travel range. This is not a CAD collision check of all components.

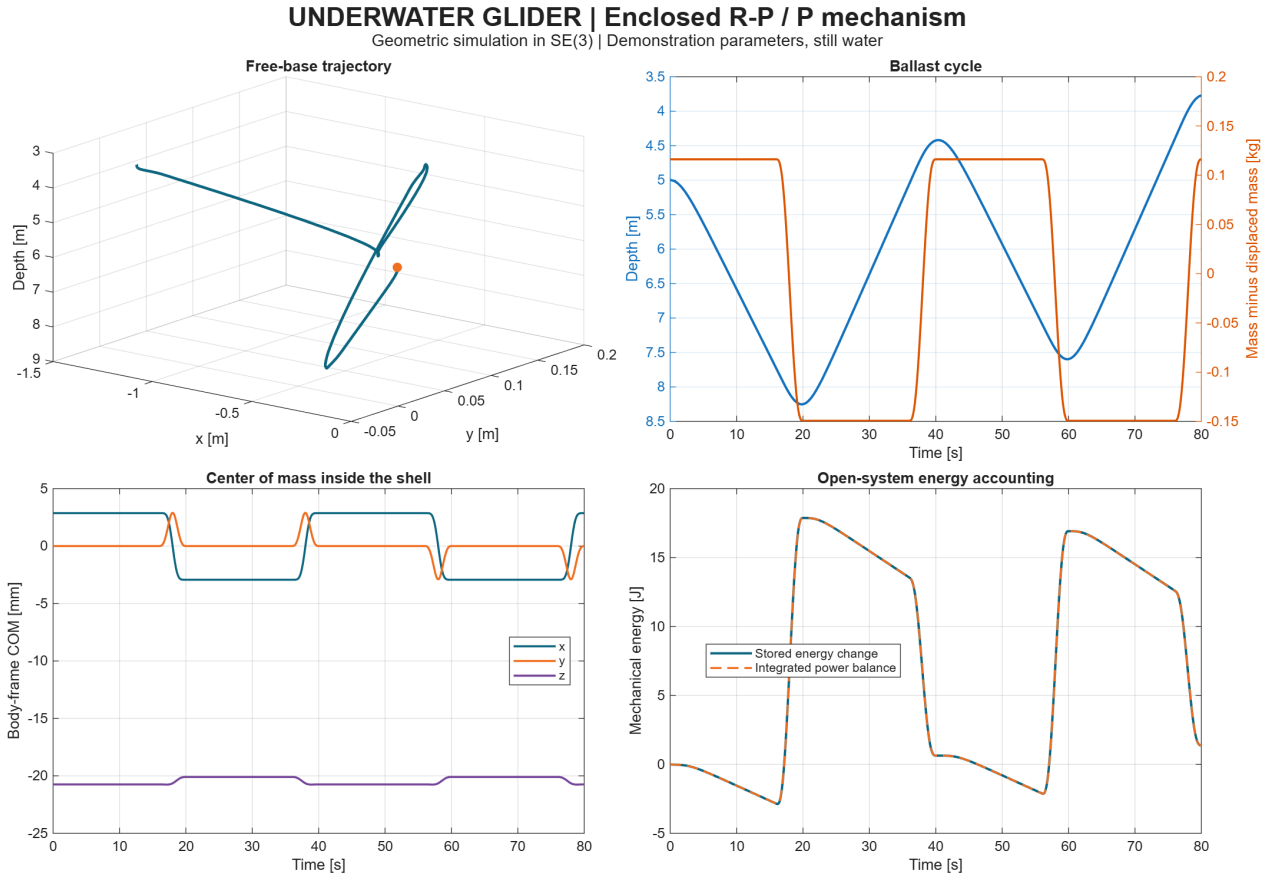
The supplied PDF also uses $d_s = .058\,\mathrm{m}$, $h = .10 - 2\theta_4$, and .05 m travel. That is a different convention. This document consistently uses (15). The file `docs/SOURCE_AUDIT.md` records the original locations of discrepancies in D, the Jacobian, buoyancy signs, inertia transformation, inertia derivatives and gravity projection. Source inputs are retained locally with hashes and excluded from distribution.

11 Verification results

Verification used MATLAB R2026a and random seed 42. The example runs for 80 seconds with a 0.02-second step and is repeated at 0.01 s.

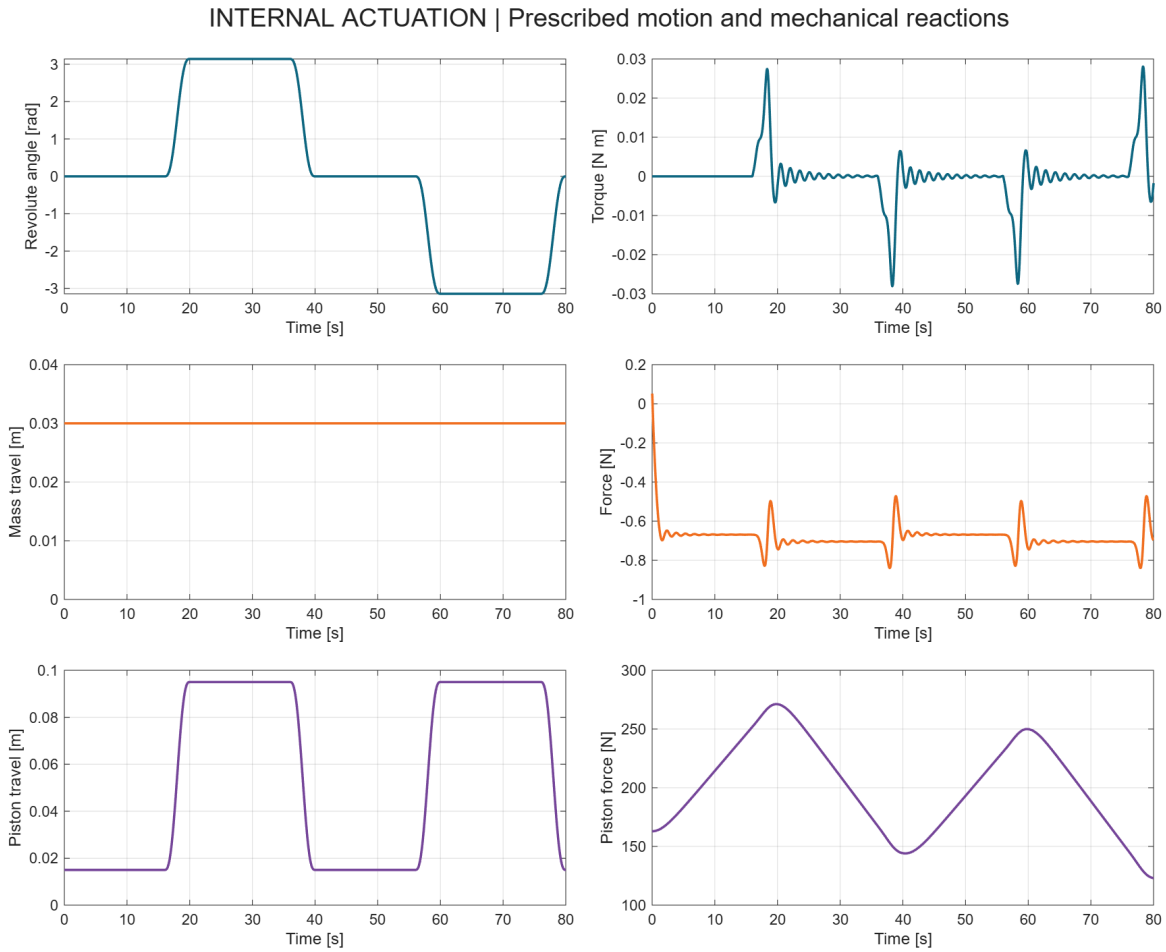
Test or indicator	Result
Random states	25
Maximum Jacobian error	4.689×10^{-10}
Relative inertia disagreement	3.290×10^{-10}
RNEA / matrix disagreement	3.730×10^{-14}
Instantaneous energy-rate error	1.349×10^{-7}
Example max $\|R^\top R - I\|_F$	1.223×10^{-14}
Position difference after refinement	1.313×10^{-5} m
Orientation difference after refinement	9.019×10^{-5} rad
Energy residual at 0.02 s	4.479×10^{-4} J
Energy residual at 0.01 s	1.082×10^{-4} J

Absolute disagreements combining different components are SI numerical consistency checks. Complete values and thresholds are available in `results/validation.json` and `tests/validate_glider.m`.

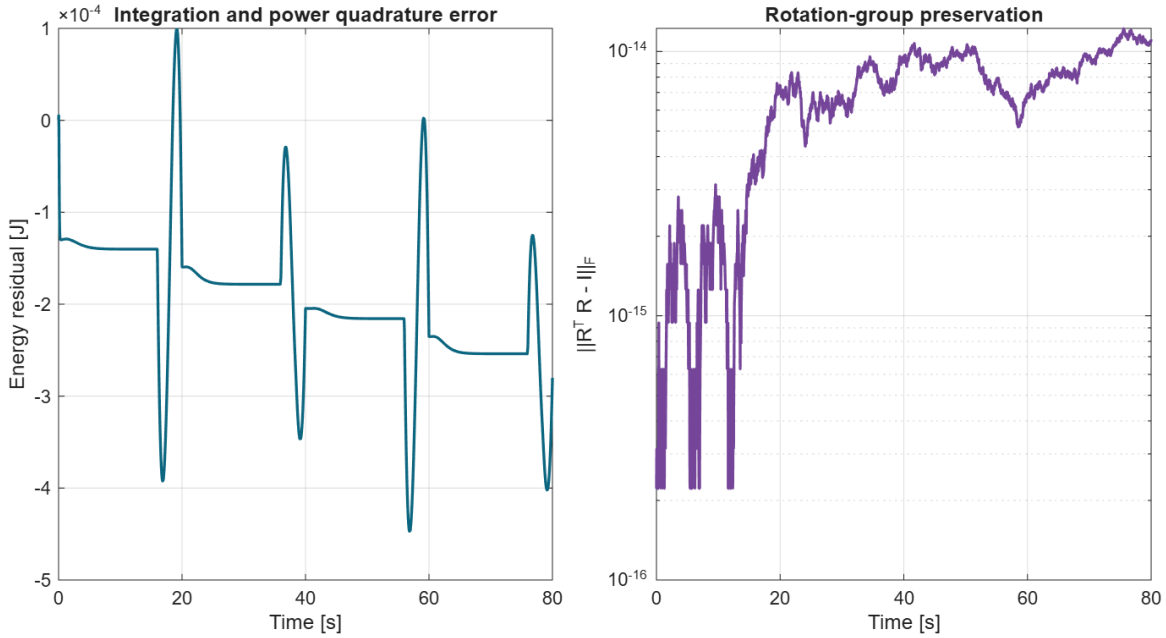


Depth ranges from 3.771 to 8.251 m, and the maximum horizontal displacement from the start is 1.387 m. These are outputs of the parameterized demonstration, not calibrated prototype predictions.

11.1 Efforts and convergence



Maximum absolute efforts are 0.02809 N m, 0.8399 N, and 271.2 N. Syringe force is dominated by pressure under the constant-back-pressure assumption. These are unsaturated model demands, not confirmed actuator capabilities.



In the isolated momentum test, reducing the step from 0.01 to 0.005 s reduces the error from 1.184×10^{-6} to 2.962×10^{-7}. The energy error falls from 7.678×10^{-9} to 1.920×10^{-9} J, consistent with second-order integration.

12 Usage and code correspondence

From the repository root:

```
setup_glider;
summary = run_glider_demo();
addpath('tests'); validate_glider;
addpath('examples');
s = load('results/demo.mat');
animate_glider(s.log, s.m);
```

Edit parameters before constructing a new model:

```
p = glider.parameters();
p.pump.back_pressure = 101325; % replace with the physical value
m = glider.model(p);
c = glider.config();
log = glider.simulate(m,c);
```

For effort-driven dynamics, select `c.mode='effort'` and assign `c.effort=@(t,H,q,nu,m) ...`, returning three efforts. Start with short horizons and a controller respecting travel and actuator limits.

Function in `+glider/`	Responsibility
`parameters, model`	Editable data and checked constants.
`kinematics`	Tree recursions, equations (5)–(6).
`spatial_inertia, water`	Spatial inertia and fill derivative.
`hydrodynamics, pump_load`	Hull loads and piston pressure.
`dynamics`	Equation (38), loads and energy.
`inverse_dynamics`	Branched RNEA with root corrections.
`hybrid_dynamics, forward_dynamics`	Free base with references or efforts.
`exp_se3, simulate`	Integration and consistent logging.
`draw_robot, plot_results`	Procedural geometry and figures.

12.1 Prototype data still required

Calibration requires CAD or measured axis and mass locations, central inertias, displaced volume, center of buoyancy, syringe pressure and flow rate, back pressure, losses, motor limits, and hydrodynamic identification. These replace the demonstration parameters while retaining the conventions and verification tests.

References

[1] K. M. Lynch and F. C. Park, *Modern Robotics: Mechanics, Planning, and Control*, Cambridge University Press, 2017. Official material, Section 8.2.

[2] T. I. Fossen, *Fossen's Marine Craft Model*. https://www.fossen.biz/html/marineCraftModel.html. Accessed October 26, 2026.

[3] J. G. Graver, *Underwater Gliders: Dynamics, Control and Design*, Ph.D. thesis, Princeton University, 2005. Full text at N. E. Leonard's laboratory.

[4] N. E. Leonard and J. G. Graver, Model-based feedback control of autonomous underwater gliders, *IEEE Journal of Oceanic Engineering*, 26(4), 633–645, 2001. https://doi.org/10.1109/48.972106.